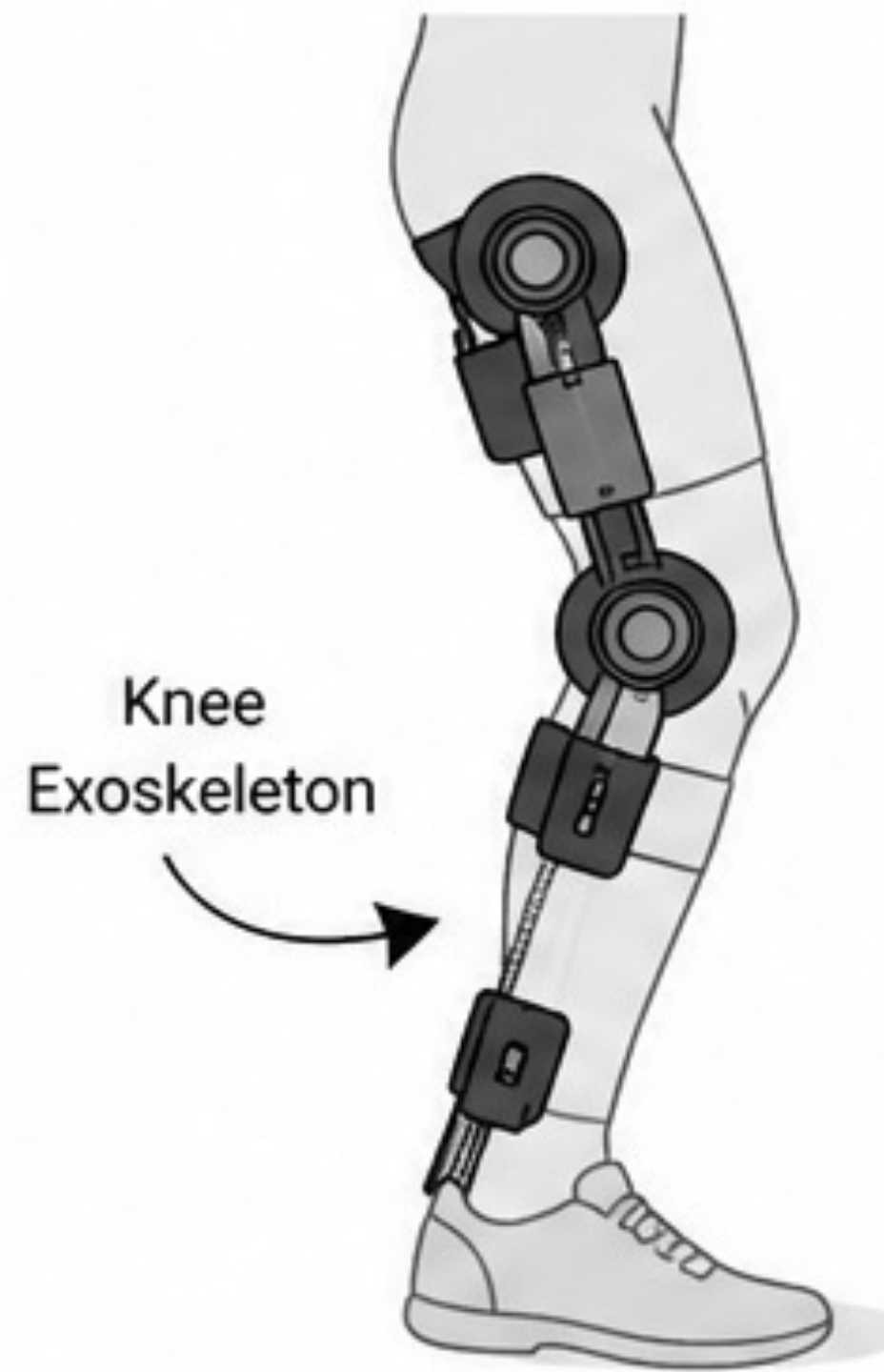




Power-based Assistance Strategy

$$P_{exo,knee} = \tau_{exo,knee} \cdot \omega_{knee}$$

$$P_{bio,knee} = M_{bio,knee} \cdot \omega_{knee}$$

Power Sign Matching

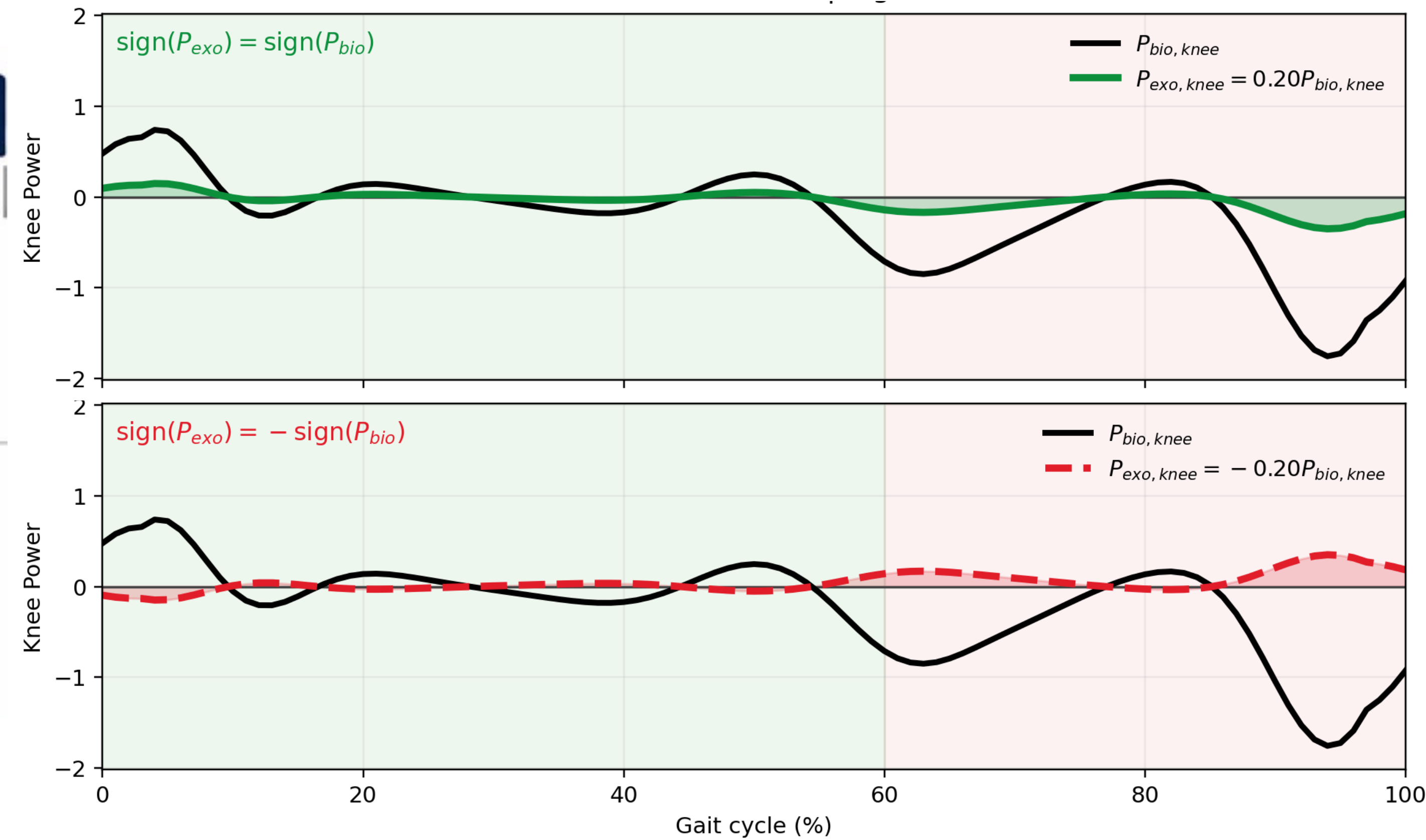
$$\text{sign}(P_{exo}) = \text{sign}(P_{bio})$$

Power Sign Mismatching

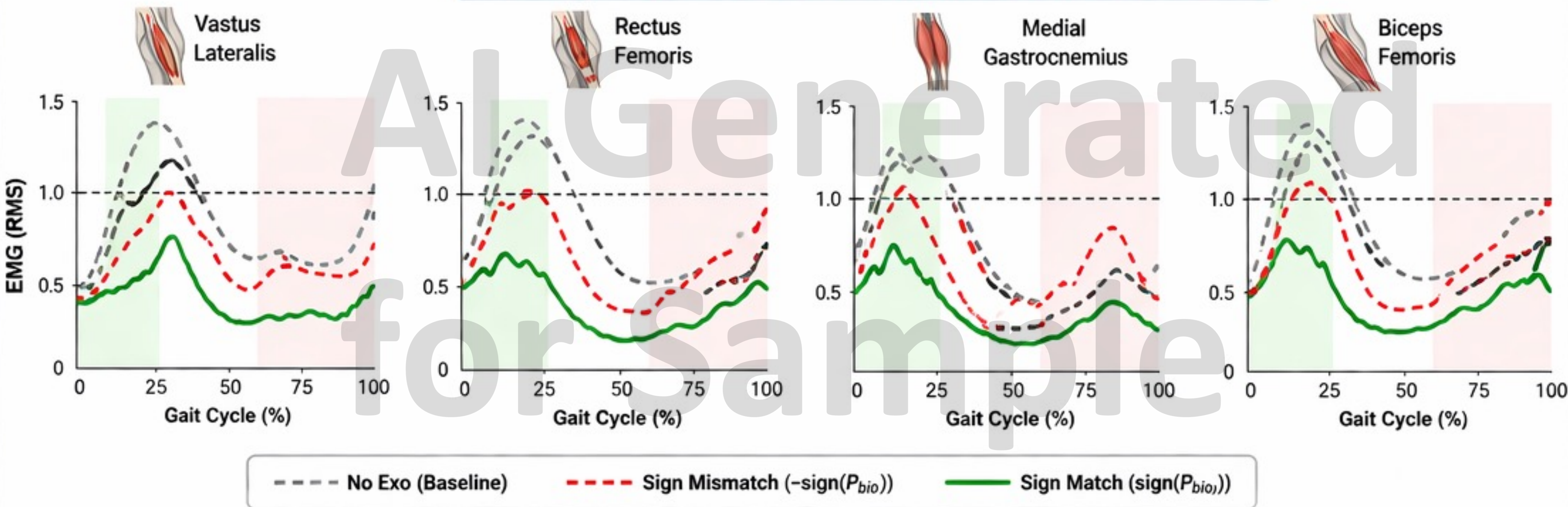
$$\text{sign}(P_{exo}) = -\text{sign}(P_{bio})$$

Power Sign Matching

$$\text{sign}(P_{exo}) = \text{sign}(P_{bio})$$



EMG (RMS) Across Gait Cycle (Normalized to Baseline)



Muscle Activity Reduction (iEMG, % vs. Baseline)

